

DISHA

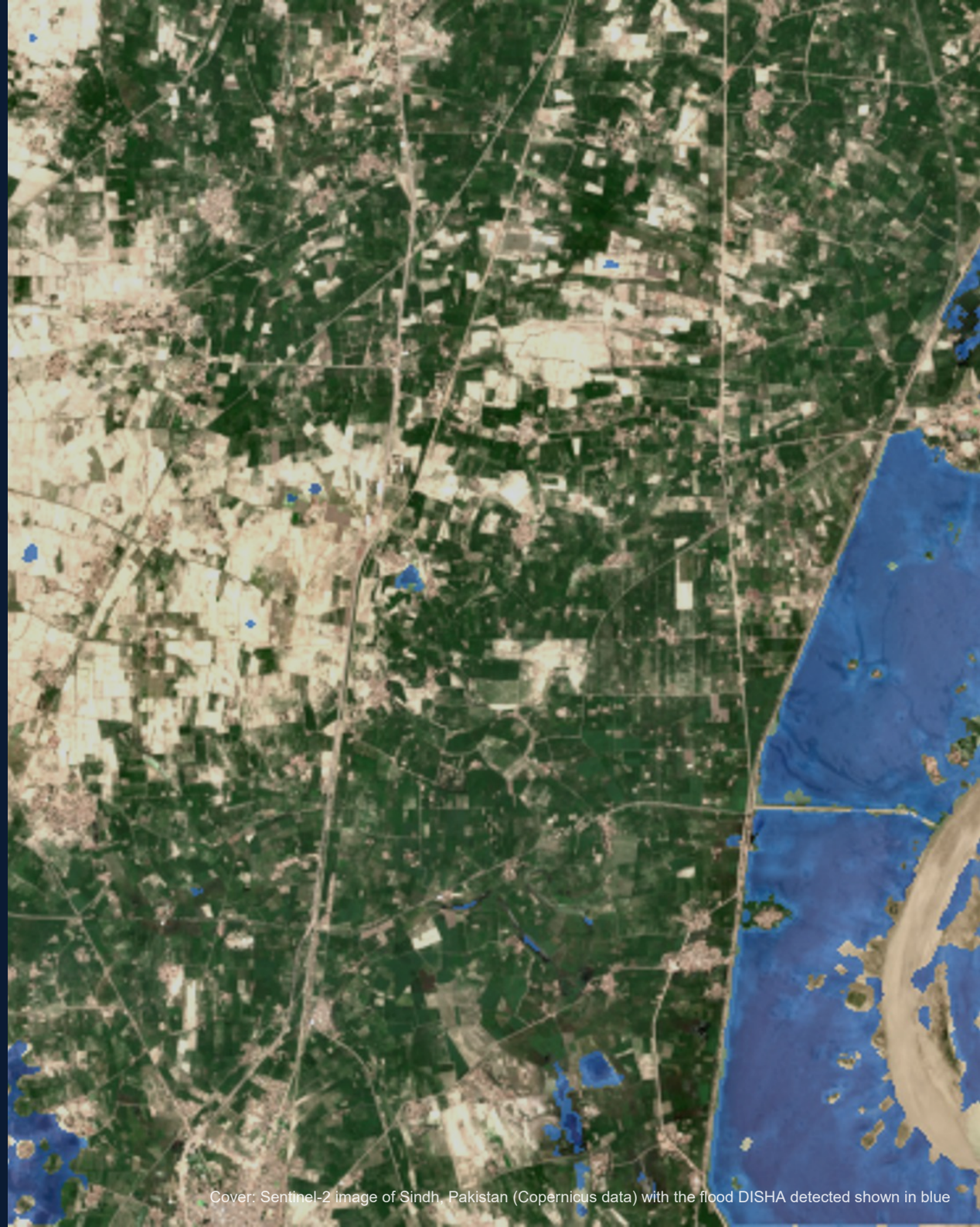
2.0

Disaster Intelligence, Situational Hazard
Assessment and Response Orchestration

**From satellite images to a clear answer
on where to respond first.**

Project guide for the team and the judges
TechFest 2026-27 - Space Technology Hackathon
Problem: Rapid post-disaster change intelligence
from multi-temporal satellite data

github.com/srushti117/Disha



Cover: Sentinel-2 image of Sindh, Pakistan (Copernicus data) with the flood DISHA detected shown in blue

The idea

DISHA in one minute

The problem. After a flood, wildfire, landslide or cyclone, response teams have hours to decide where to go first. The information they need is scattered: satellite images, rainfall, population, roads, hospitals, field reports. A satellite change map shows *what changed*. It does not say *who needs help first*.

What DISHA does. It takes satellite images from before and after a disaster, finds where the damage is, links it to people, hospitals and roads, and produces a ranked, explained list of places to reach first. It then proposes routes, shelters and resources, and recalculates every time a responder reports from the ground.

The question DISHA answers:

Where should emergency responders go first, what should they do there, which resources should be deployed, which route should they take, and why?



The five questions it answers

Question	Where DISHA answers it
1. What happened?	Satellite change detection (radar first, optical when clear)
2. Who is affected?	Population, vulnerability, hospitals, schools, roads
3. Where do we go first?	Priority score P1 to P4 with the reasons shown
4. What should we do?	Recommended actions, safest route, shelters, resource plan
5. Did the situation change?	New satellite passes and field reports recalculate everything

Why it matters

The gap DISHA is built to close

The figures below come from the team's Round 1 proposal (sources cited there: NDMA and the Geological Survey of India).

12%

of India's land is flood-prone (~40 Mha)

5,700 km

of 7,516 km coastline exposed to cyclones

12.6%

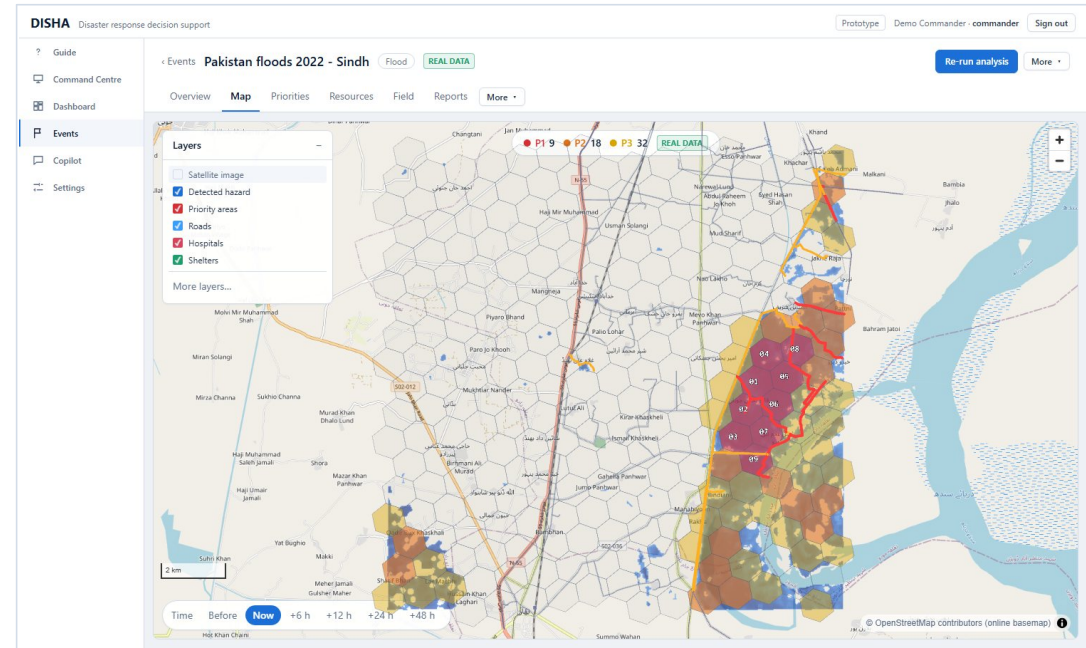
of land area is landslide-prone

72 hours

golden window to save lives

What goes wrong today

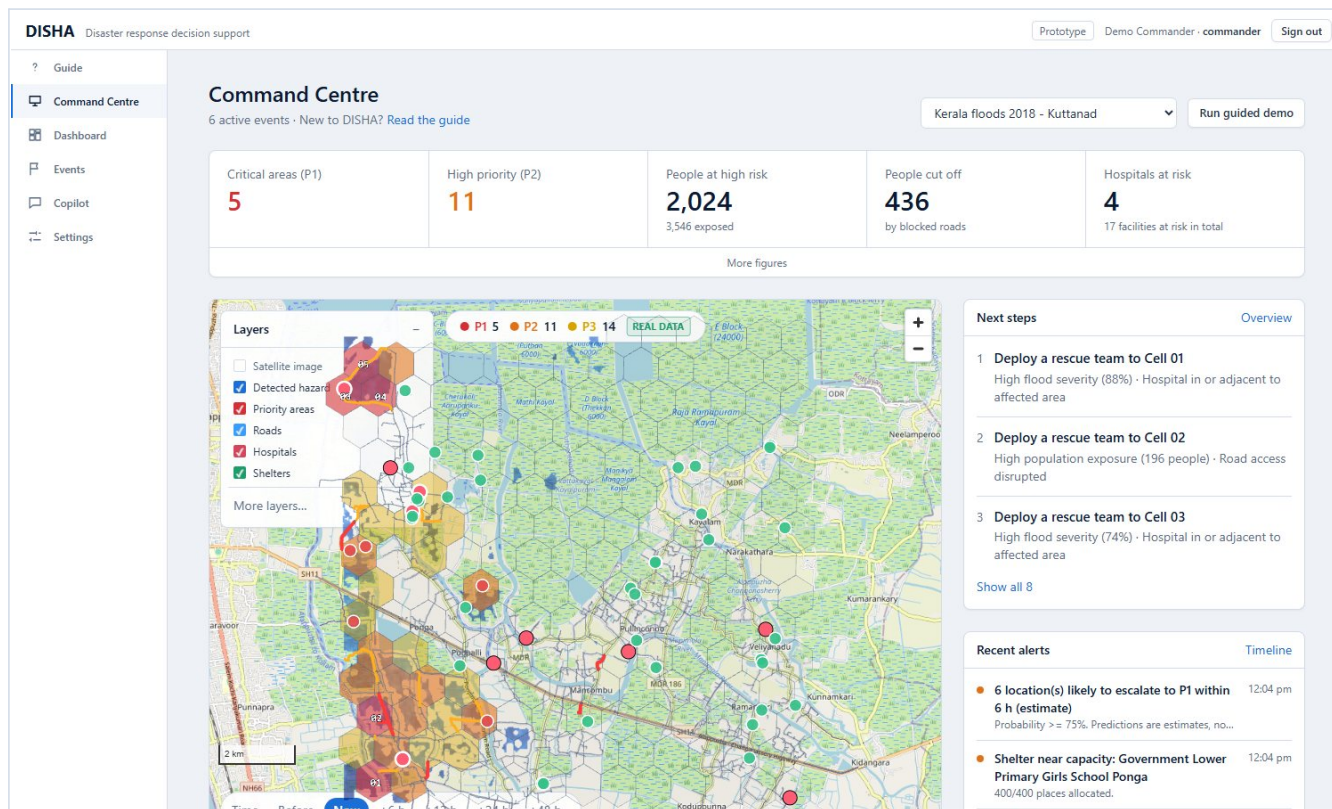
- **Too slow.** Manual interpretation of satellite images can take days, after the golden window has closed.
- **Blind under cloud.** Optical satellites cannot see through monsoon and cyclone cloud, exactly when the disaster happens.
- **Change is not a decision.** A change map does not say which village to reach first.



Real example: detected flooding (blue) and ranked hexagon cells (red = P1) next to the Indus river, Sindh, Pakistan, after the 2022 floods. Real Sentinel data.

The product

The Command Centre: the first screen

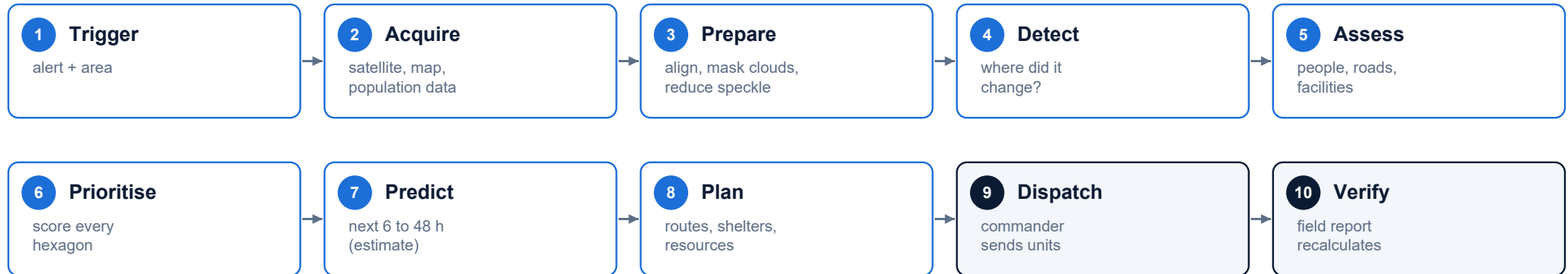


What you are looking at

- **Five headline numbers** at the top: critical areas, high-priority areas, people at high risk, people cut off by blocked roads, hospitals at risk. A single click on *More figures* shows five more.
- **The map** shows the detected hazard, the priority hexagons, roads, hospitals and shelters. Click any hexagon to see why it has its priority.
- **Next steps** and **Recent alerts** on the right tell the commander what to do now.
- **Run guided demo** plays the whole workflow in about a minute.
- A simple **Guide** page explains the project in plain English.

How it works

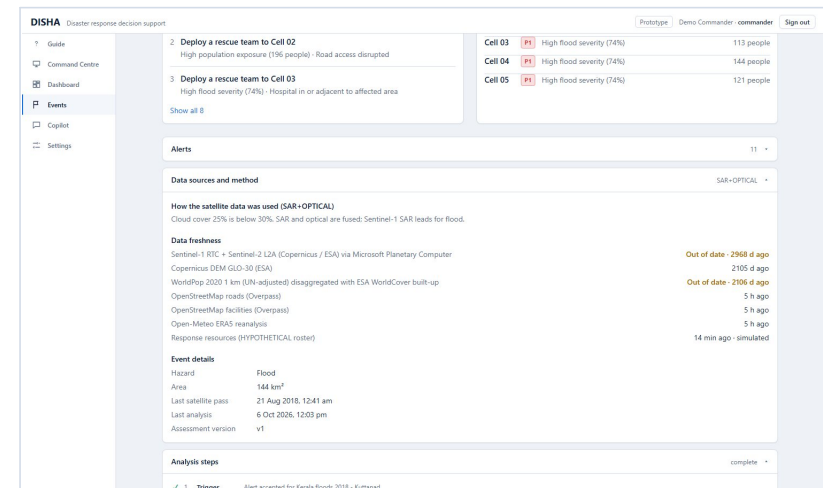
What happens when you press Run analysis



Steps 1-8 run automatically in the background. Steps 9-10 are people: a commander dispatches, a responder verifies.

In plain words

- **Acquire** downloads the satellite scenes, terrain, land cover, population, roads and rainfall for the chosen area.
- **Detect** compares before and after radar images and marks where water, burn scars or ground change appear.
- **Assess** divides the area into hexagons about 500 m across and works out people, facilities and road access for each.
- **Prioritise** scores every hexagon from P1 (critical) to P4 (low) and records the reasons.
- **Plan** prepares routes to the top priority areas and a shelter plan; a commander then decides on resources and dispatch.

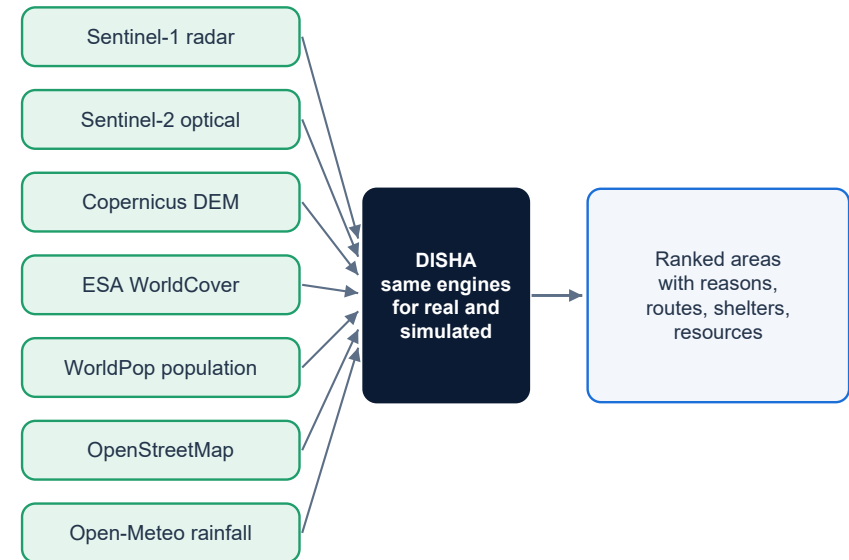


Data

Real data and simulated data: what is what

DISHA runs in two modes and always labels which one you are looking at. **Real data** events download actual open datasets when you run them. **Simulated** events generate synthetic imagery and facilities so the product can be shown offline; the algorithms are identical.

Input	Source	Status
Radar before/after	Sentinel-1 RTC (Copernicus / ESA), Microsoft Planetary Computer	Real
Optical image	Sentinel-2 L2A, cloud-masked	Real (optional)
Terrain	Copernicus DEM GLO-30	Real
Land cover	ESA WorldCover 2021	Real
Population	WorldPop 2020 (1 km) spread onto built-up land	Real, modelled estimate
Roads, hospitals, schools, bridges, places	OpenStreetMap (Overpass)	Real
Rainfall	Open-Meteo (ERA5 for past events)	Real
Rescue units	No real source	Hypothetical roster
Shelter capacity	OSM has none; assumed by type	Assumed
Children / elderly shares	Not available	Skipped, not invented



The screenshot shows the DISHA web interface. On the left is a sidebar with navigation links: Guide, Command Centre, Dashboard, Events (selected), Caplist, and Settings. The main content area is titled 'New event' and includes a form to define the hazard and area of interest (AOI). Below the form, there's a section 'Area of interest' with tabs for 'Demo event', 'Draw polygon', 'Coordinates', 'Admin area', and 'Upload GeoJSON'. Under the 'Demo event' tab, several scenarios are listed, each with a radio button to select it. The scenarios are labeled as either 'Real data' or 'Simulated'.

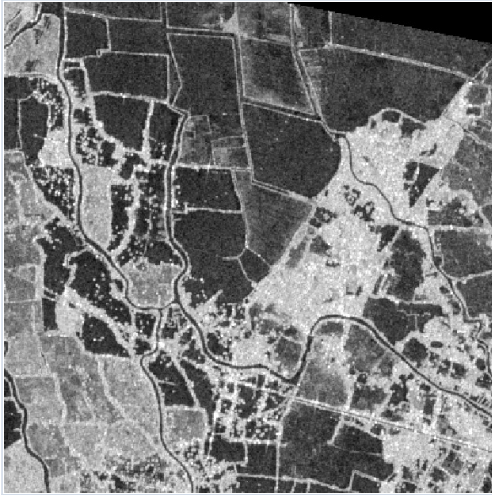
Scenario	Label
Kerala floods 2018 - Kuttanad	Real data
Kerala floods 2018 - Chalakudy	Real data
Pakistan floods 2022 - South	Real data
Kerala floods 2018	Simulated
Wayanad Landslide 2024	Simulated
Synthetic Urban Flood	Simulated
Forest Wildfire	Simulated
Coastal Cyclone	Simulated

Scenarios are labelled Real data or Simulated when you start an event.

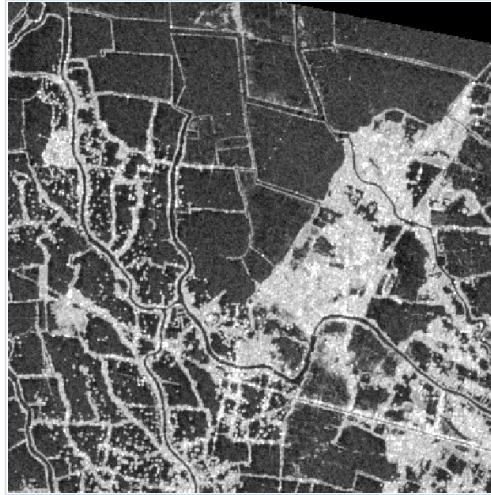
Real data in action

Kerala floods, August 2018: the Kuttanad backwaters

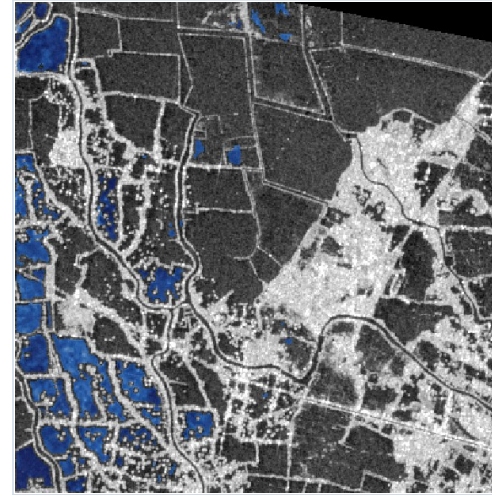
Two real Sentinel-1 radar scenes from the **same orbit track** (9 August and 21 August 2018) plus a real Sentinel-2 image. Water is dark in radar, so a sudden drop in brightness marks new flooding. The permanent backwaters are excluded.



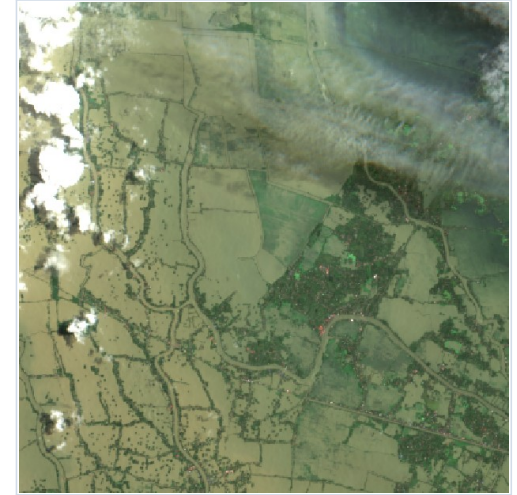
Radar before (Sentinel-1)



Radar after (Sentinel-1)



Detected flood (blue) on the radar



Satellite image after (Sentinel-2)

8.3 km² newly flooded area detected	25.4 km² permanent water (excluded)	189 hexagon cells analysed	P1 5 P2 11 critical and high-priority cells	2,024 people at high risk (modelled)	1,430 real road segments checked
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Numbers come from running the real pipeline on these scenes. Population is a modelled estimate from WorldPop; detection has not been validated against ground truth.

How to read the images

- **Dark areas in radar** are smooth surfaces such as open water. Bright areas are towns and vegetation.
- **Blue overlay** marks pixels that became dark between the two dates and passed the checks (not permanent water, not steep ground).

Why this event

- Kuttanad lies below sea level and stayed under water for weeks, so the radar pass on 21 August still shows the flood.
- It also has dense real roads and 140 mapped bridges, so blocked-road analysis is meaningful.

Method

How the flood is found in the radar image

- **Same view, twice.** Only scenes from the same satellite track and direction are compared, so differences are real, not geometry.
- **Reduce noise.** Radar images are grainy (speckle); they are averaged in linear power before comparing.
- **Look for drops.** The log-ratio (after minus before, in dB) is thresholded automatically with Otsu's method, with limits that keep the threshold physically sensible.
- **Remove false alarms.** Permanent water (already dark before), steep slopes (radar shadow) and tiny blobs are masked out.
- **Use optical when it helps.** On cloud-free pixels, a Sentinel-2 water index confirms or weakens the radar result.
- **Score it.** Each pixel gets a severity index and a confidence value; both are carried into the priority score.

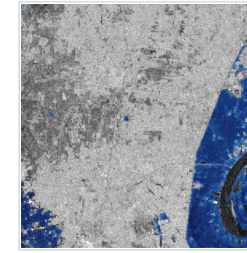
Honest limits. Radar misses flooded built-up areas and flooded vegetation, and the severity is an intensity index, not water depth in metres. Detection has not been checked against surveyed flood maps.



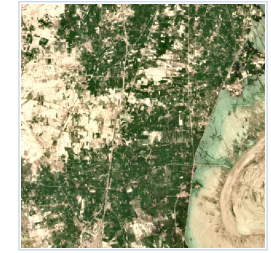
Radar before
(Sentinel-1)



Radar after (Sentinel-1)



Detected flood (blue) on
the radar



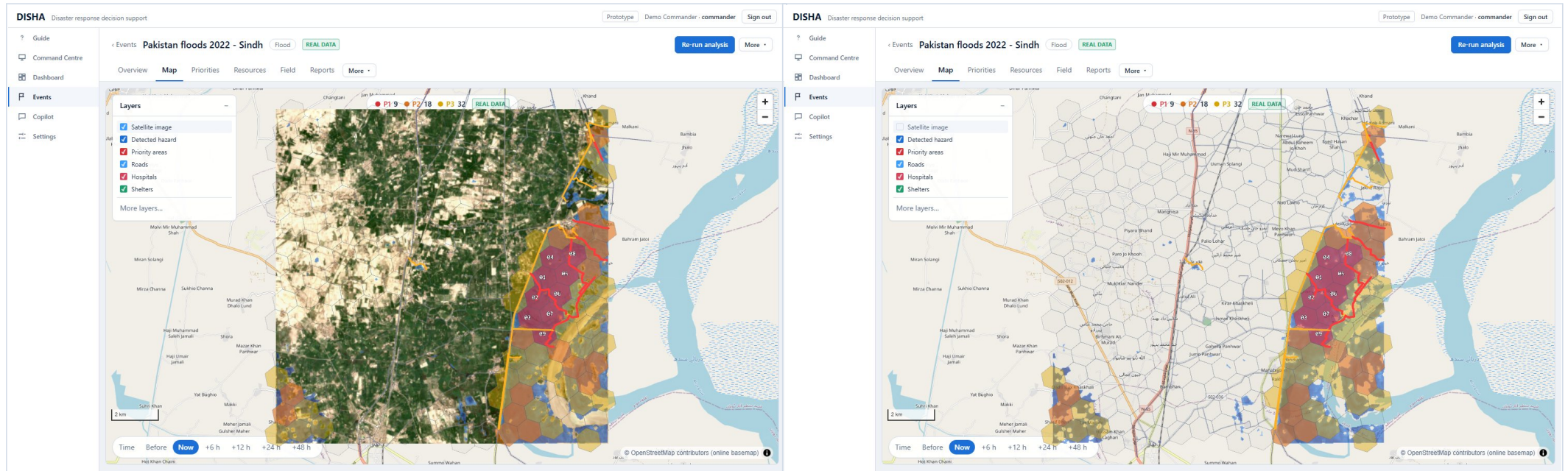
Satellite image after
(Sentinel-2)

Second real example: Sindh, Pakistan, 2022. Left to right: radar before, radar after, detected flood, satellite image.

The system explains its sensor choice. Example from the app: *Optical imagery downgraded due to high cloud cover. Sentinel-1 SAR selected as primary source. With no usable optical scene it says so and runs radar-only.*

Real data in action

Pakistan floods 2022: Sindh, beside the Indus



Sentinel-2 satellite image with DISHA's priority hexagons.

Same area on the street map: detected flood (blue), blocked roads (red), P1 cells numbered.

33.5 km²

flooded area detected

P1 9 P2 18

critical and high cells

5,436

people at high risk

4,215

people cut off by blocked roads

266

cells analysed

What the commander sees

- Detected flood runs along the Indus side of the area, so P1 cells cluster there.
- Roads that cross the flooded strip are marked blocked, which cuts off several villages.

What to treat with care

- Population is a modelled estimate; actual counts may differ.
- Only 273 road segments are mapped here, so isolation may be understated or overstated where OpenStreetMap is sparse.

Honesty

What happens when the data is weak

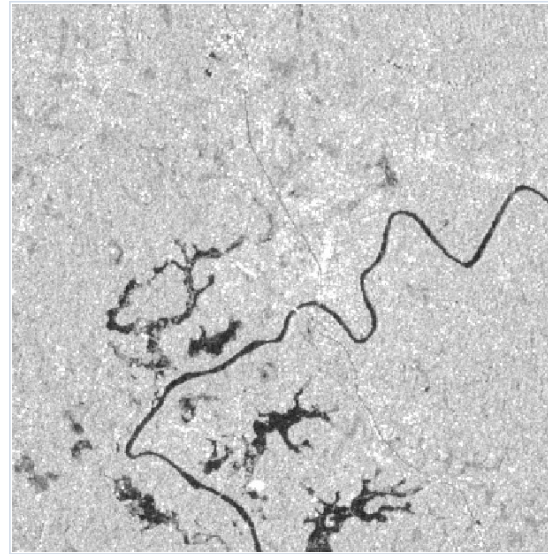
A decision tool must be clear about what it cannot see. Two real examples from this project:

Chalakyudy, Kerala (Aug 2018). The first usable radar pass after the flood peak is 21 August, when most water had already receded. DISHA detects only **0.44 km²** and ranks **no P1 areas**. We kept the scenario to show that the tool reports what the satellite shows, not what we expect.

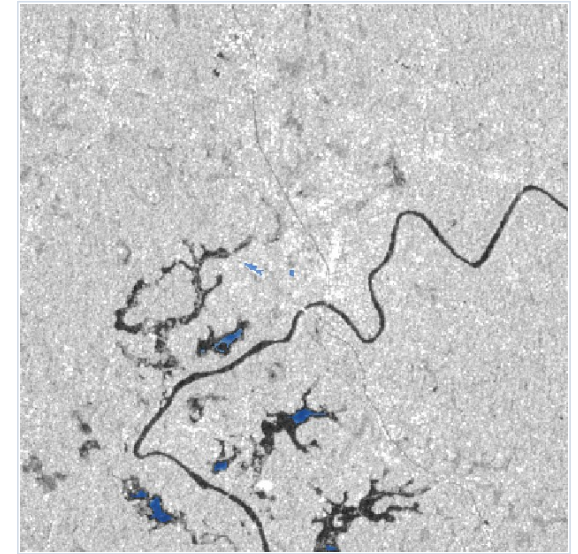
Wayanad landslide (Jul 2024). With the only same-track scene available (6 August) and 30 m radar, the real change signal was too weak and the rule detected nothing. A silent miss on a real disaster would mislead, so this scenario was **removed** rather than shown.

Low-confidence labels. Every area shows a confidence level (high, medium, low, insufficient data). When satellite coverage, population or roads are weak, confidence drops and the app says verify on the ground first.

Principle: the app never replaces missing data with invented values. Missing demographic factors are skipped, unavailable population is shown as unknown, and failed downloads stop the job with a readable error.



Chalakyudy radar after the event



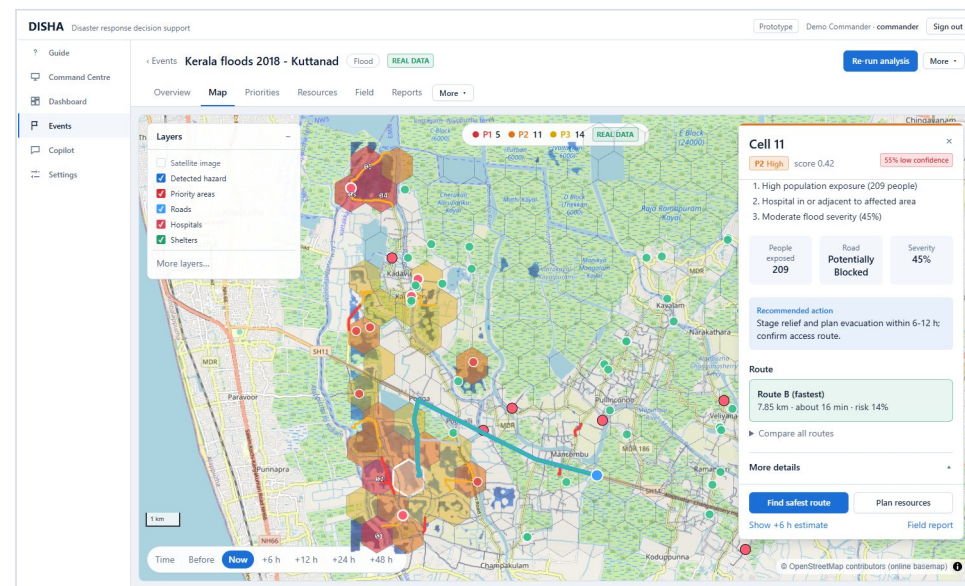
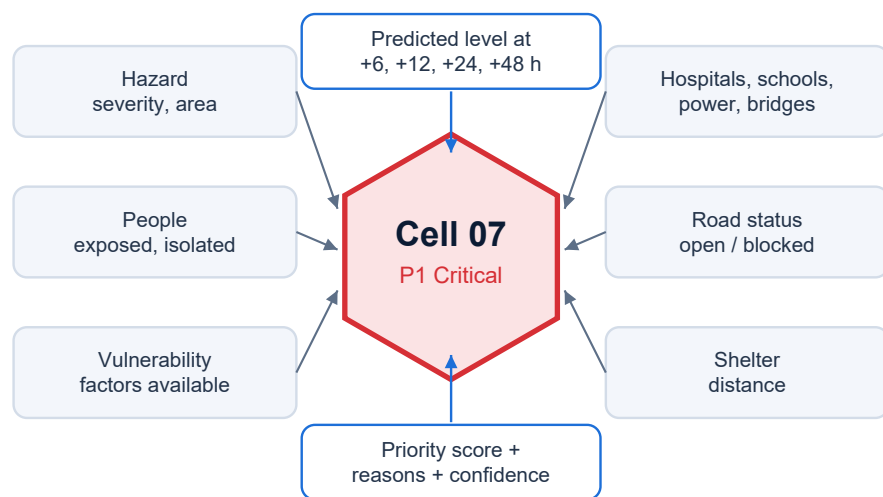
Detected flood: only small patches

Digital twin

From pixels to places: one hexagon, one live state

The area is divided into H3 hexagons about 500 m across (resolution is configurable). Each hexagon is a small **digital twin**: its hazard, people, facilities, road access, priority and predicted future are stored together and recalculated whenever new information arrives (a new satellite pass, a field report, or a change in settings).

Every recalculation is saved as a numbered assessment, so DISHA can show **what changed** between versions and raise an alert such as *Cell 12 escalated from P2 to P1 with the reasons*.

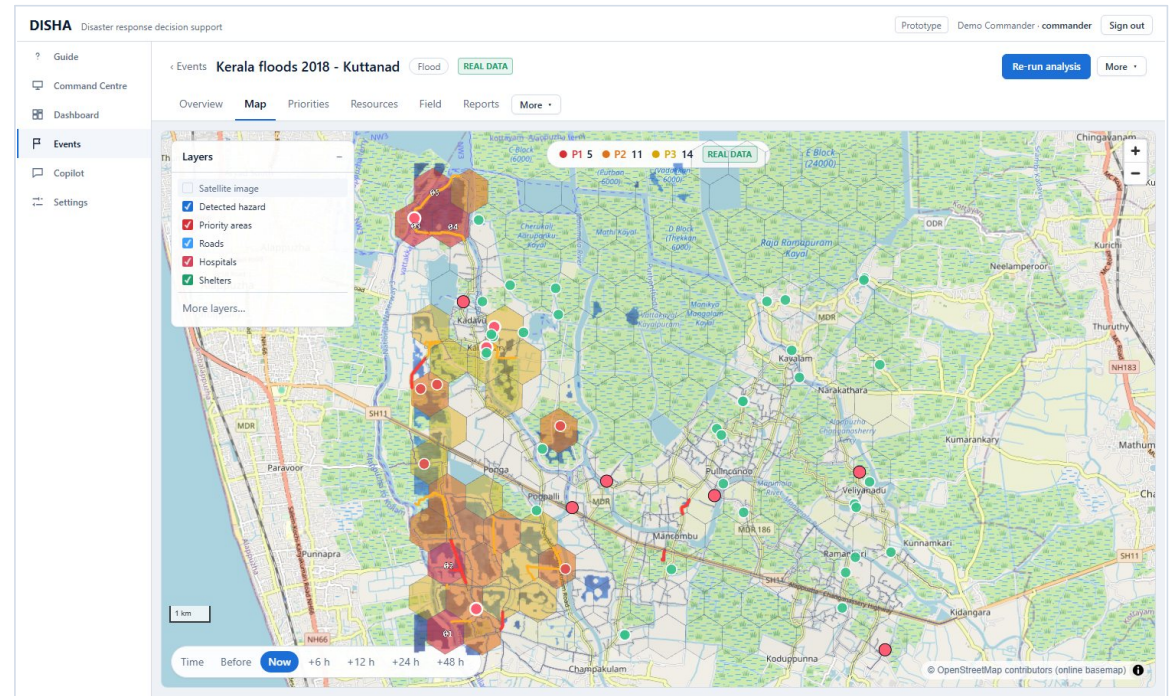


The cell panel: key facts first, details one click away.

Impact assessment

From change to consequences

- **People.** Pixel-level population (WorldPop 1 km spread onto built-up land) summed per hexagon: total, exposed to the hazard, at high risk, and potentially cut off.
- **Vulnerability.** A configurable score from density, distance to a hospital, isolation, road access and severity. Age and housing factors are skipped when no data exists.
- **Critical facilities.** Hospitals, schools, power, water, fire and police, bridges; each carries a risk and the reason.
- **Roads.** Every real OpenStreetMap road segment is checked against the hazard and labelled open, potentially blocked, blocked or unknown. Bridges and single points of failure are marked critical.
- **Isolation.** Blocked roads are removed from the network; hexagons that can no longer be reached from a response base count as cut off.
- **Knock-on risk.** Dependency chains: a flooded substation puts its hospital at risk; a blocked bridge isolates a village and delays rescue.



Real roads, hospitals (red dots) and shelters (green dots) over the detected flood and priority hexagons.

Prioritisation

The priority score: simple enough to explain



P1 Critical

score 0.48 and above

P2 High

score 0.34 to 0.48

P3 Moderate

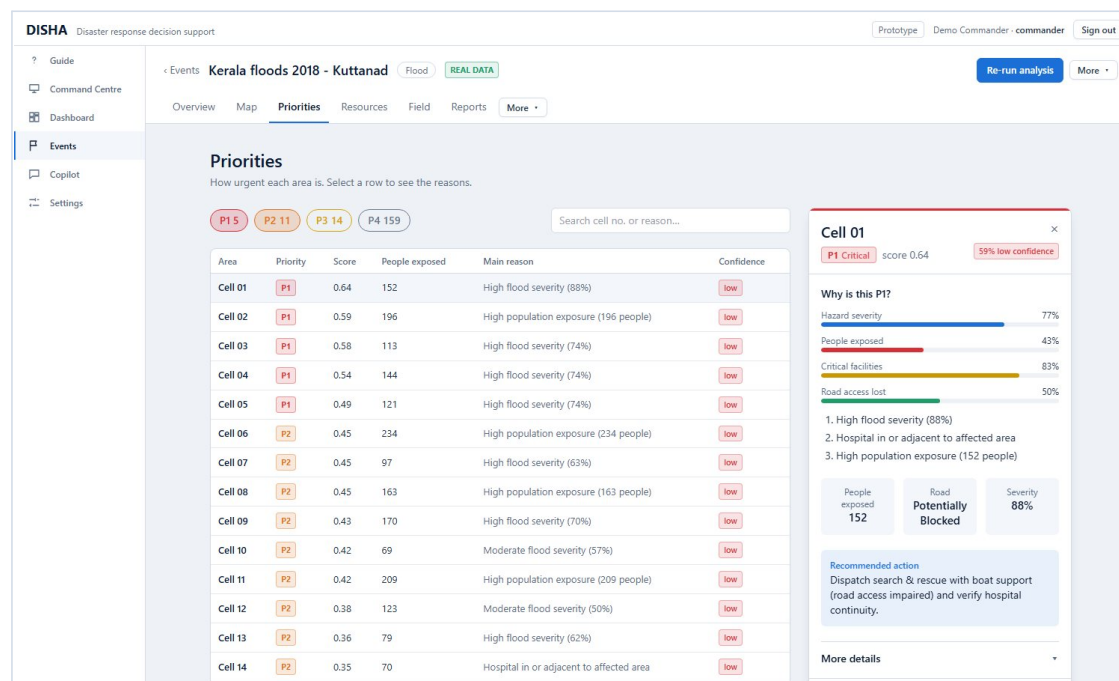
score 0.20 to 0.34

P4 Low

score below 0.20

Why a formula, not a black box

- The weights come from the original DISHA proposal (35/30/20/15) and can be changed on the Priorities page; changing them recalculates everything and is recorded in the audit log.
- Each part is scaled between 0 and 1 and multiplied by its weight, so the four contributions always add up exactly to the score.
- Areas with almost no people (under 10) and no critical assets are capped below P2: an empty burning forest is flagged, but rescue teams are not sent there.
- Severity is multiplied by detection confidence, so a doubtful detection ranks lower.

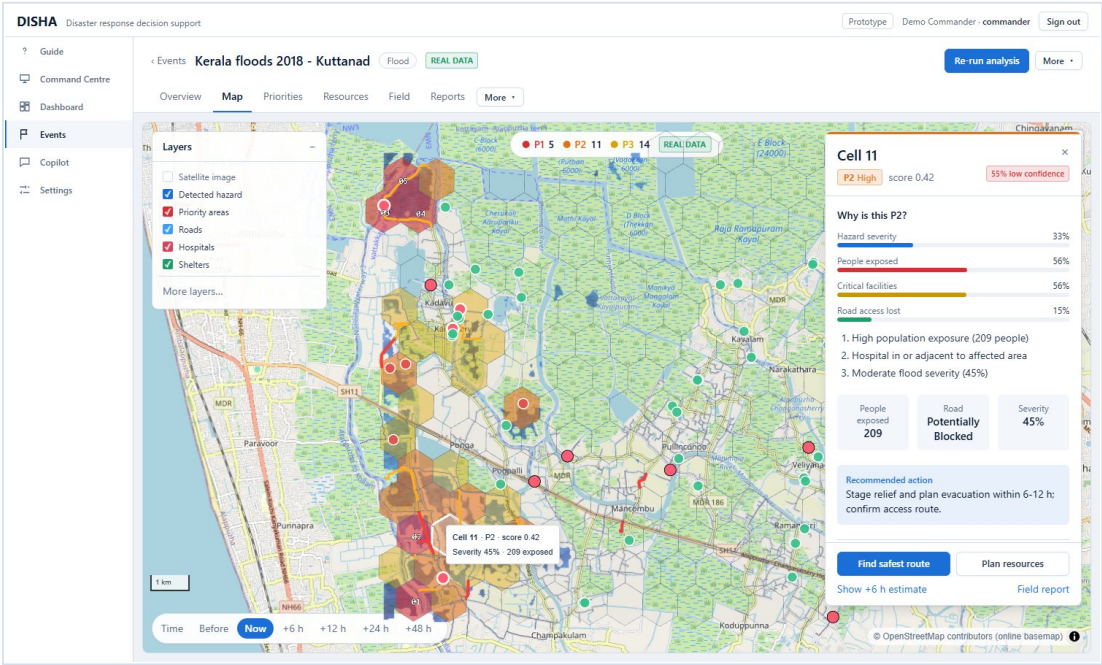


Explainability

Why is this P1? Every ranking shows its reasons

- **Four bars** show how much hazard, people, facilities and lost road access contribute.
- **Plain reasons** in order of importance, for example: *High population exposure (209 people); Hospital in or adjacent to affected area; Moderate flood severity (45%)*.
- **Recommended action** written from the data, for example *Stage relief and plan evacuation within 6-12 h; confirm access route*.
- **Confidence** combines satellite, population, roads, weather and facilities, weighted by quality and freshness. Stale data lowers it.
- **Sources and age** of each input are listed under *More details*.

Confidence label	Meaning
High (80% and above)	Act on it; routine checks
Medium (60% to 80%)	Likely correct; verify when possible
Low (40% to 60%)	Verify on the ground first
Insufficient data	Do not rely on it



Selecting a hexagon opens the explanation panel.

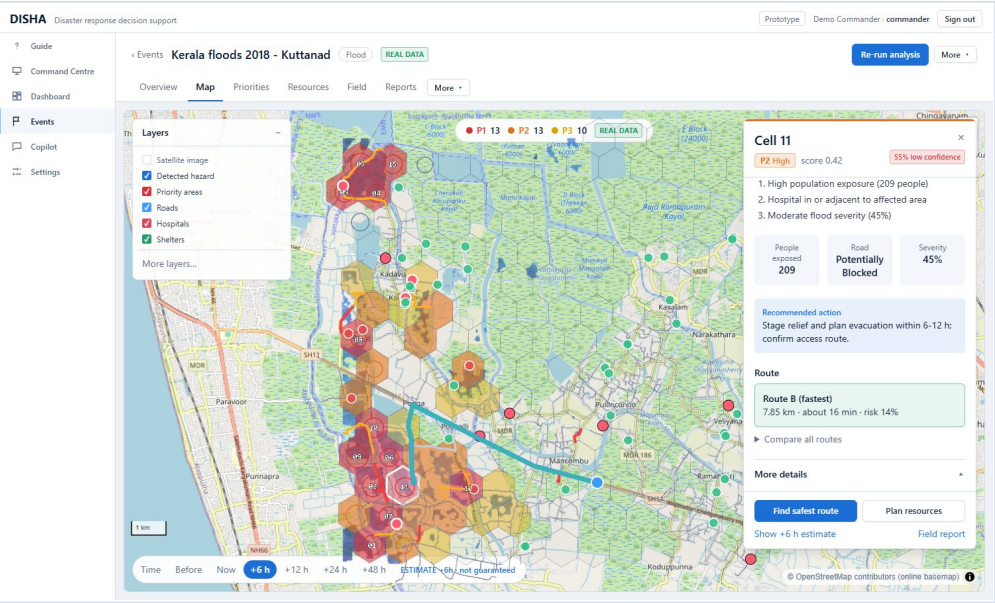
Looking ahead

Predictions and the time control

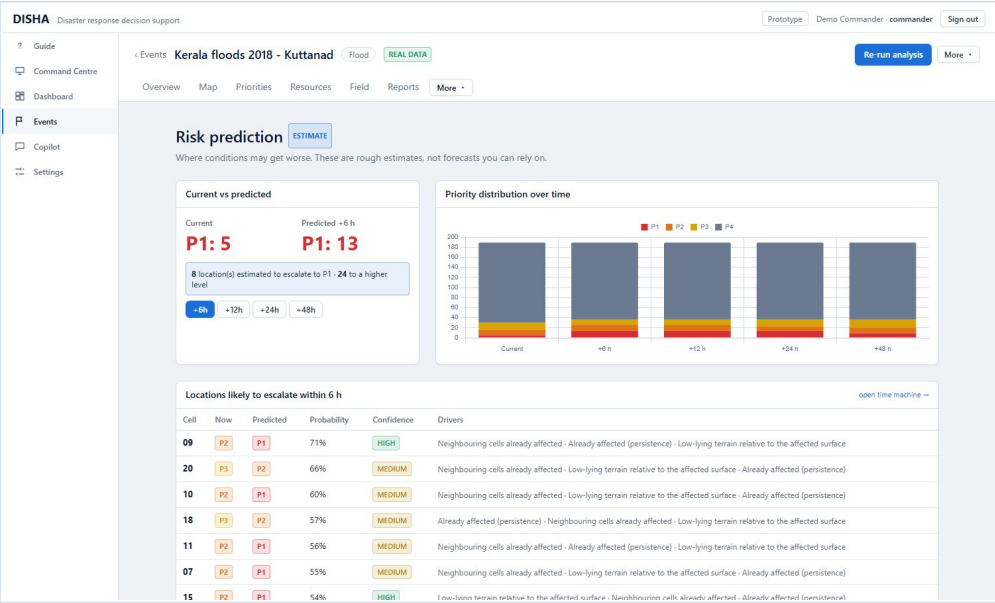
DISHA estimates which areas may get worse in the next 6, 12, 24 and 48 hours. A transparent model combines how much of the surrounding area is already affected, how low the terrain is, the rainfall that follows, and persistence. Each estimate has a probability, and its confidence decays with time.

The control at the bottom of the map switches between *Before*, *Now* and the four forecast horizons so commanders can see the likely spread.

Always an estimate. The coefficients are set by hand, not trained on observed floods, and have not been validated. Every prediction is labelled ESTIMATE and never shown as a guarantee.



The map at +6 hours: estimated spread (labelled as an estimate).



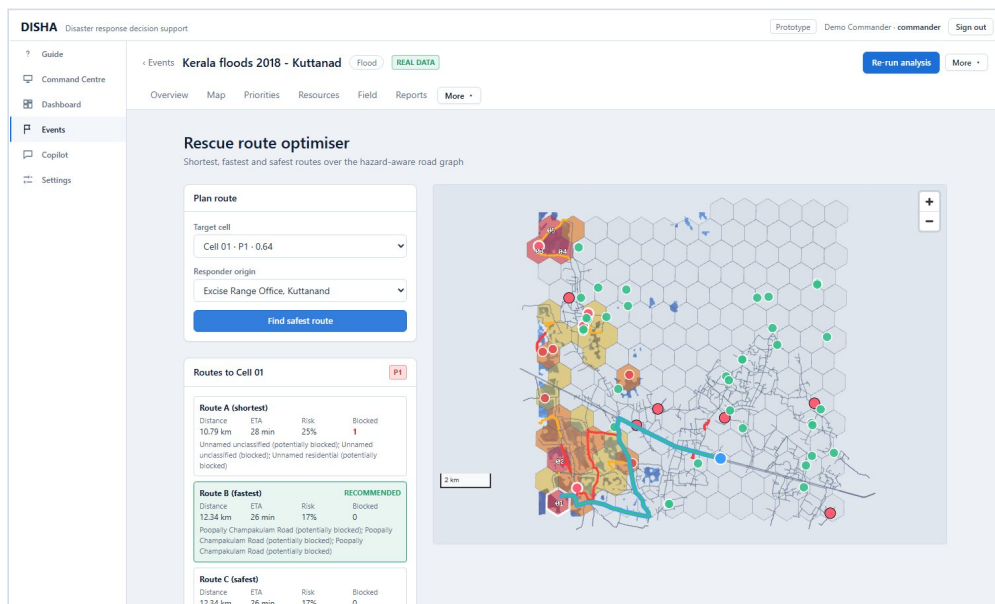
Predictions page: current counts against predicted counts for each horizon.

Response planning

Which road should responders take?

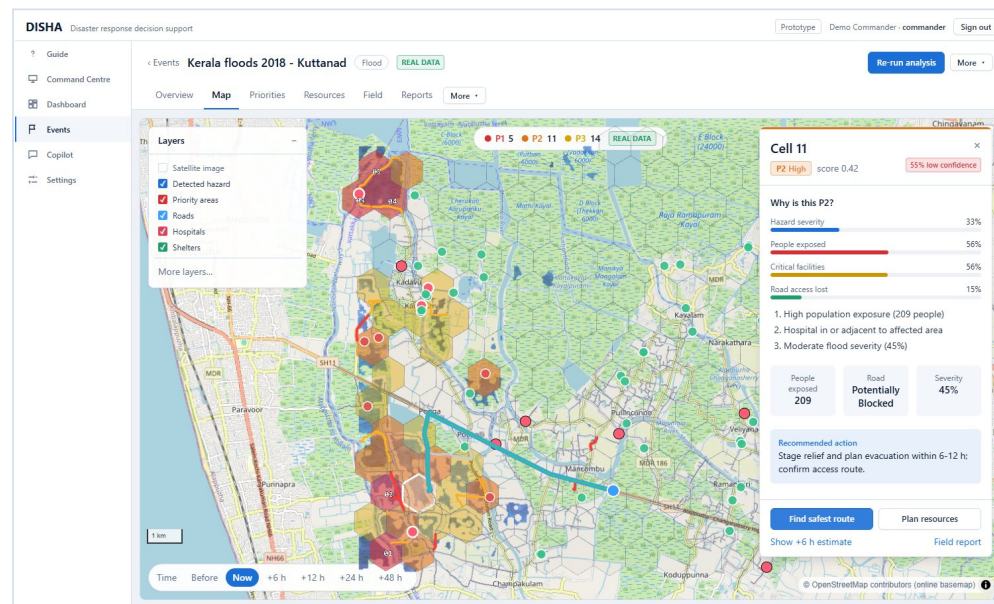
DISHA builds a road network from OpenStreetMap, marks each segment by the hazard, and computes three routes from a response base to a priority area:

- **Shortest** by distance. If it crosses a blocked segment, it is flagged.
- **Fastest** by estimated travel time, avoiding blocked roads.
- **Safest** minimising risk, also avoiding blocked roads.



Route comparison: recommended route drawn on the real road network.

- A recommendation weighs time and risk. If no fully open road exists, it says so and points to boat or air access.
- When the last stretch lies inside the flooded zone, DISHA warns that the final approach needs a boat or a high-clearance vehicle.
- Travel times are estimates, not guarantees.



Route from the cell panel on the map: Find safest route.

Response planning

Shelters and resources

Shelters. People in P1 and P2 hexagons are assigned to the nearest safe shelter reachable by open roads, respecting capacity. The planner flags unreachable shelters, shelters inside the hazard zone, overcapacity and any shortfall.

In real-data events the rescue-unit roster is **hypothetical**: no real resource feed exists. It is flagged on screen and in exports.

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Evacuation planner

People in P1/P2 cells are assigned to safe, road-reachable shelters with remaining capacity

Need shelter2,143

Placed1,947

Shortfall196

Flags (10)

UNREACHABLE

No safe shelter reachable by open road from cell 2; 196 people need air/boat evacuation.

UNSAFE SHELTER

GNS LP SCHOOL, NEDUMUDY lies inside the hazard zone and is excluded.

OVERCAPACITY

Unnamed school is at 100% of capacity.

UNSAFE SHELTER

ANGANAVADY lies inside the hazard zone and is excluded.

UNSAFE SHELTER

ANGANAVADY lies inside the hazard zone and is excluded.

UNSAFE SHELTER

ANGANAVADY lies inside the hazard zone and is excluded.

OVERCAPACITY

St. Mary's High School Kainakary is at 100% of capacity.

OVERCAPACITY

Holy Family Girl's High School Kainakary is at 100% of capacity.

OVERCAPACITY

Government Lower Primary Girls School Ponga is at 100% of capacity.

UNSAFE SHELTER

Government Upper Primary School Thottuvathala lies inside the hazard zone and is excluded.

Assignments

Cell	Shelter	People	Road dist.	Capacity left
01 P1	Government Lower Primary Girls School Ponga	152	8.25 km	248
03 P1	St. Mary's High School Kainakary	113	6.77 km	287

Shelter capacity

Little Flower Girls High School, Pulinkunnu	0 / 400
Fr. Thomas Ponakara Central School	0 / 400
SH UP School kannady	0 / 400
GNS LP SCHOOL, NEDUMUDY Δ in hazard zone	0 / 400
Govt.Technical High School Kavalam	0 / 400

Evacuation planner: who goes to which shelter, and remaining capacity.

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Resources

Decide which teams, boats and ambulances go where

The units here are hypothetical: there is no real resource feed. Replace them with real unit data before operational use.

Plan and dispatch

Plan resources

Dispatch

Release all

Free units: 12 rescue teams, 4 boats, 6 ambulances, 4 medical teams, 3 drones, 6 relief vehicles

Critical areas without a team5 of 5

Critical areas covered80%4 of 5

Slowest first arrival32.4 min estimate

Not enough units for: Cell 04 (drone), Cell 05 (boat), Cell 05 (medical team), Cell 05 (drone), Cell 05 (drone)

Assignments (27)

Unit	Area	Arrives in	Status
Rescue Team 01	Cell 01	27 min	Planned
Boat 01	Cell 01	32 min	Planned
Ambulance 01	Cell 01	21 min	Planned
Medical Team 01	Cell 01	23 min	Planned
Drone 01	Cell 01	7 min	Planned

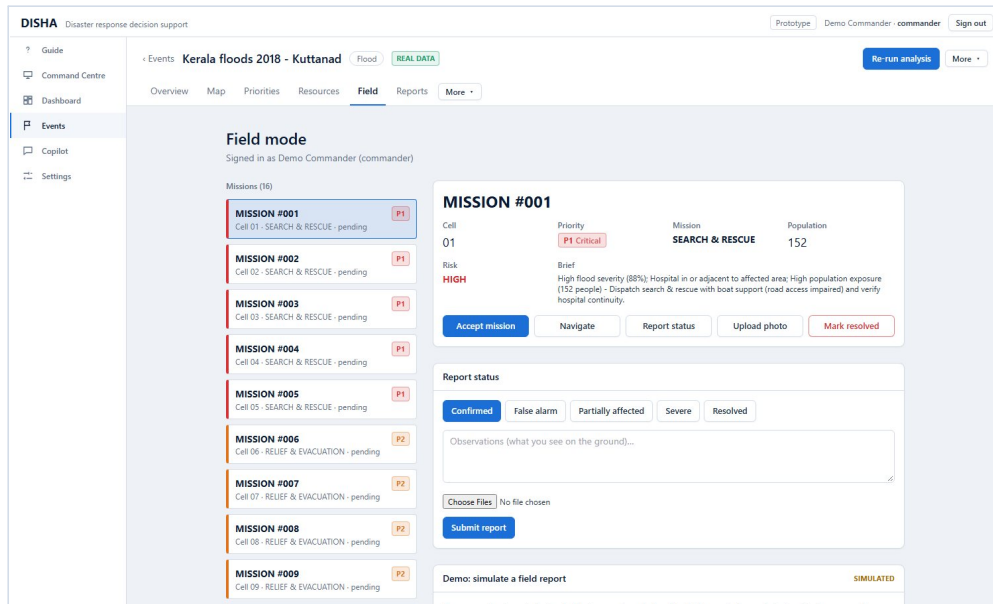
Resource plan: each unit with its reason and arrival time.

Closing the loop

Field verification: the part that makes it a loop

A responder opens their mission, reports what they see and may upload a photo. DISHA recalculates immediately.

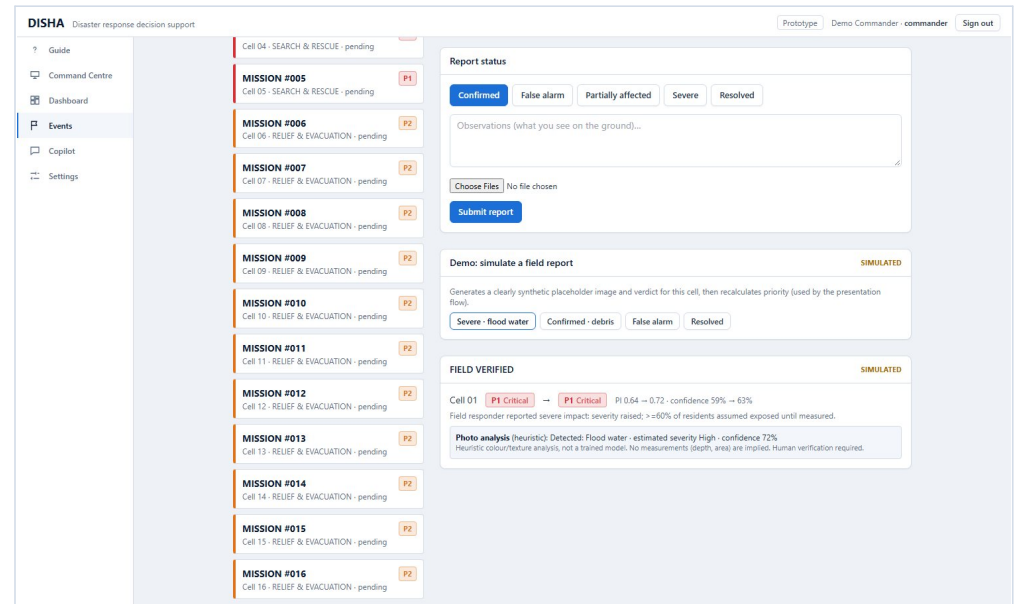
- **Confirmed:** confidence rises.
- **False alarm:** severity and exposure drop sharply; the label is stored for model feedback.
- **Partially affected:** scaled down.



The responder's missions (one per priority area).

- **Severe:** severity is raised and a minimum share of residents assumed exposed until measured.
- **Resolved:** the area drops out of the response list.
- Reports are stored as labelled observations that can be exported for future model training.

Photo analysis is a simple colour and texture heuristic, not a trained model; it states a limited confidence and gives no measurements.



A verified report (a simulated one here) changes the priority.

Closing the loop

After the report: dispatch and what changed

The commander dispatches the planned units; after a field report the overview shows **what changed** between assessments, and alerts such as *Cell NN escalated from P2 to P1* appear with their reasons. Everything is also written to the incident timeline and audit log.

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Resources

Decide which teams, boats and ambulances go where

The units here are hypothetical: there is no real resource feed. Replace them with real unit data before operational use.

Plan and dispatch

Plan resources Dispatch Release all

Free units: 7 rescue teams, 0 boats, 0 ambulances, 0 medical teams, 0 drones, 1 relief vehicle

Critical areas without a team 0 of 5

Critical areas covered 80% 4 of 5

Slowest first arrival 32.4 min estimate

Not enough units for: Cell 04 (drone), Cell 05 (boat), Cell 05 (medical team), Cell 05 (drone), Cell 06 (drone)

Assignments (27)

Unit	Area	Arrives in	Status
Rescue Team 01	Cell 01	27 min	Dispatched
Rescue Team -> Cell 1 (P1): P1 with 152 people exposed. Unit 1/1; nearest available (ETA 27 min).			
Boat 01	Cell 01	32 min	Dispatched
Boat -> Cell 1 (P1): road access lost / community isolated. Unit 1/1; nearest available (ETA 32 min).			
Ambulance 01	Cell 01	21 min	Dispatched
Ambulance -> Cell 1 (P1): hospital at risk. Unit 1/1; nearest available (ETA 21 min).			
Medical Team 01	Cell 01	23 min	Dispatched
Medical Team -> Cell 1 (P1): vulnerable population / hospital continuity. Unit 1/1; nearest available (ETA 23 min).			
Drone 01	Cell 01	7 min	Not dispatched

Dispatched units after pressing Dispatch.

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High priority (P2) 11

People at high risk 2,024 3,589 exposed

People cut off 436 by blocked roads

Hospitals at risk 4 17 facilities at risk in total

More figures

Situation summary

Current analysis (assessment v2) identifies 5 P1 and 11 P2 cells across the affected region (7.6 km2 affected, SAR+OPTICAL sensor mode). The highest-priority location, Cell 01, scores 0.72: high flood severity (90%); high population exposure (195 people); hospital in or adjacent to affected area. 2,024 people are at high risk and 3,589 are exposed; 436 are potentially isolated by blocked roads. 17 critical facilities are at risk, including 4 hospital(s). 8 road segments are blocked and 30 potentially blocked. Estimate: 8 additional location(s) may escalate to P1 within 6 hours (heuristic model, not guaranteed). 1 field report(s) received.

Recommended next steps Plan resources

1 Deploy Rescue Team 01 to Cell 01 High flood severity (90%) - High population exposure (195 people)

2 Deploy Rescue Team 03 to Cell 02 High population exposure (196 people) - Road access disrupted

3 Deploy Rescue Team 05 to Cell 03 High flood severity (74%) - Hospital in or adjacent to affected area

Show all 8

Most urgent areas All priorities

Cell 01 P1 High flood severity (90%) 195 people

Cell 02 P1 High population exposure (196 people) 196 people

Cell 03 P1 High flood severity (74%) 113 people

Cell 04 P1 High flood severity (74%) 144 people

Cell 05 P1 High flood severity (74%) 121 people

What changed since the last assessment

Overview after a field report: what changed since the last assessment.

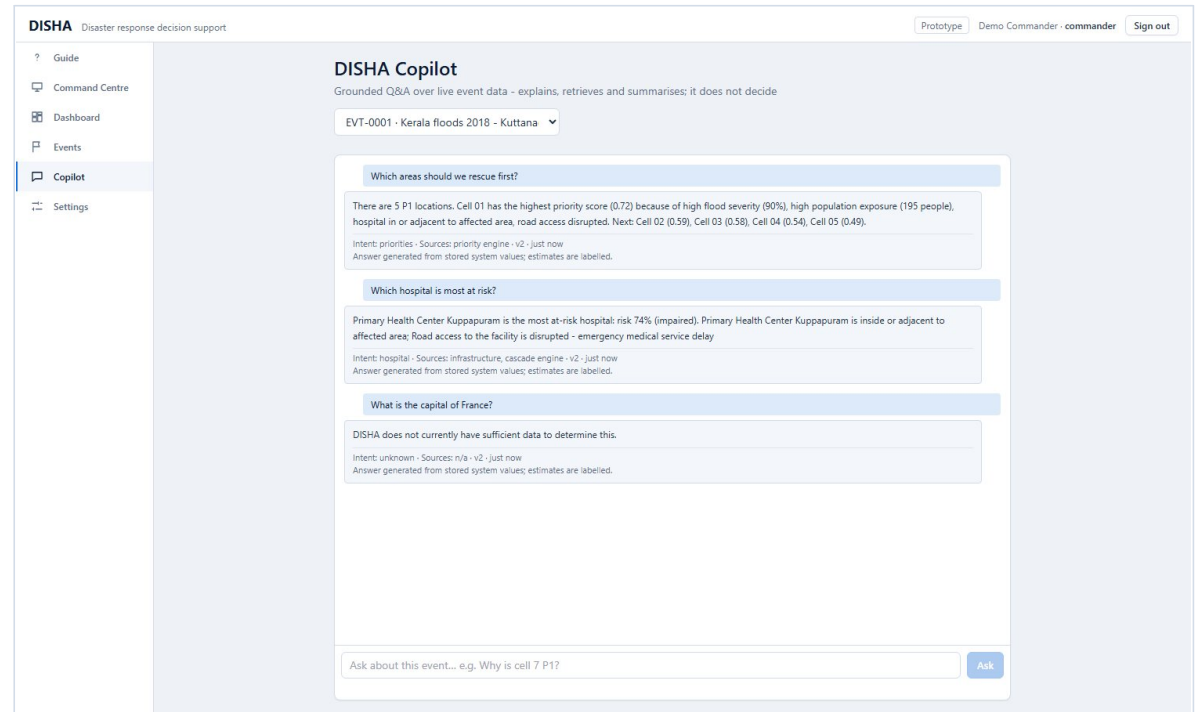
Assistant

DISHA Copilot: answers from the data, never guesses

The Copilot is **not** a general chatbot. It recognises the question, runs the matching query on the event's database (priorities, routes, predictions, resources, data health) and writes the answer from those values, with the sources and assessment version shown.

If the information is not there it replies exactly: *DISHA does not currently have sufficient data to determine this.* (see the last question in the screenshot).

- Which areas should we rescue first?
- Why is Cell 12 P1?
- Give me the safest route to Cell 12.
- How many people are currently at high risk?
- Which hospital is most at risk?
- What changed since the previous assessment?

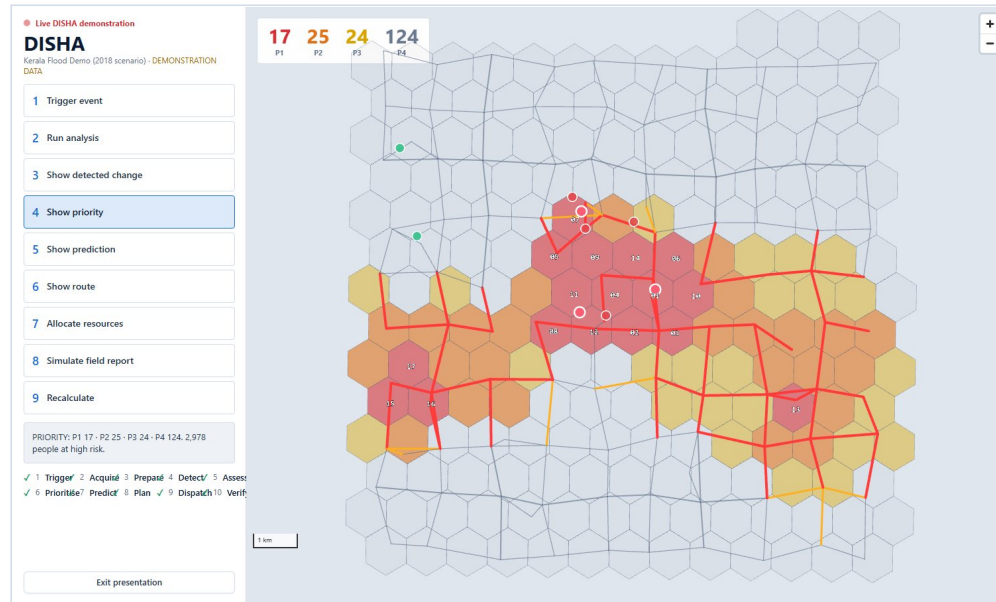


Grounded answers with their sources, and a refusal when data is missing.

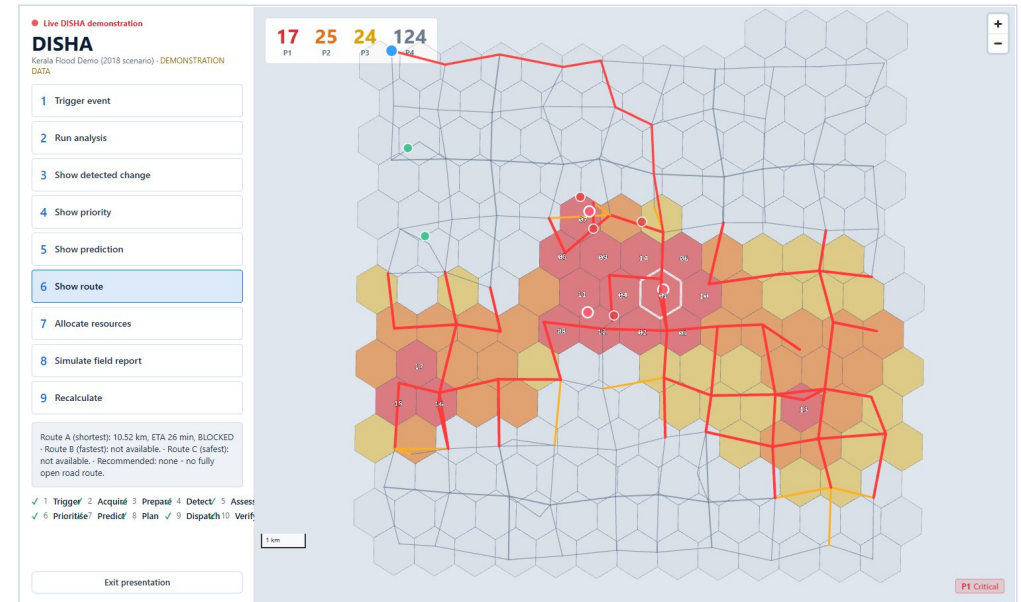
Demonstration

Presentation mode and the guided demo

Presentation mode is a projector-friendly screen with nine big steps: trigger the event, run the analysis, show the detected change, show priorities, show the prediction, show the route, allocate resources, simulate a field report, recalculate. **Run guided demo** on the Command Centre plays the whole workflow in about 45 seconds with narration, using simulated data so it works without internet.



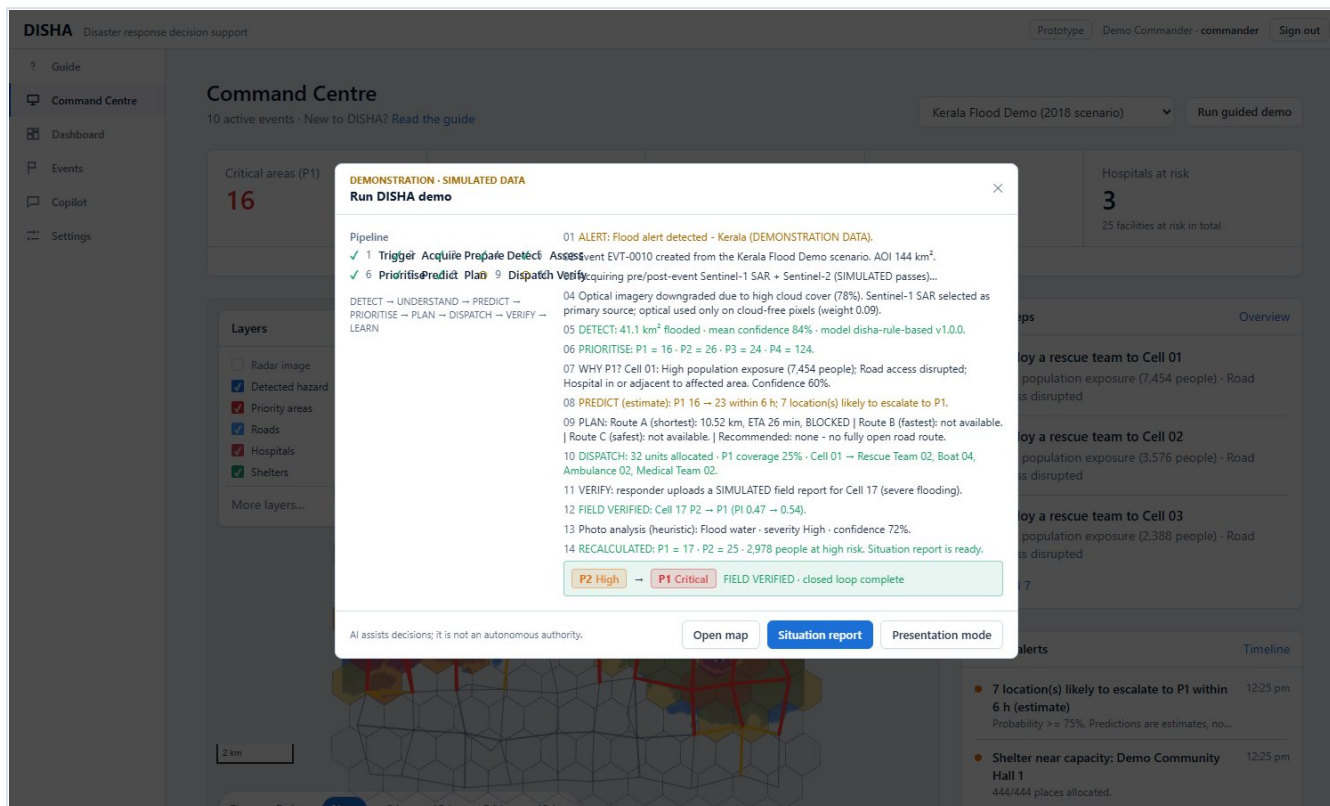
Presentation mode, step 4: priorities (simulated Kerala scenario).



Presentation mode, step 6: route to the top priority area.

Demonstration

The guided demo, finished

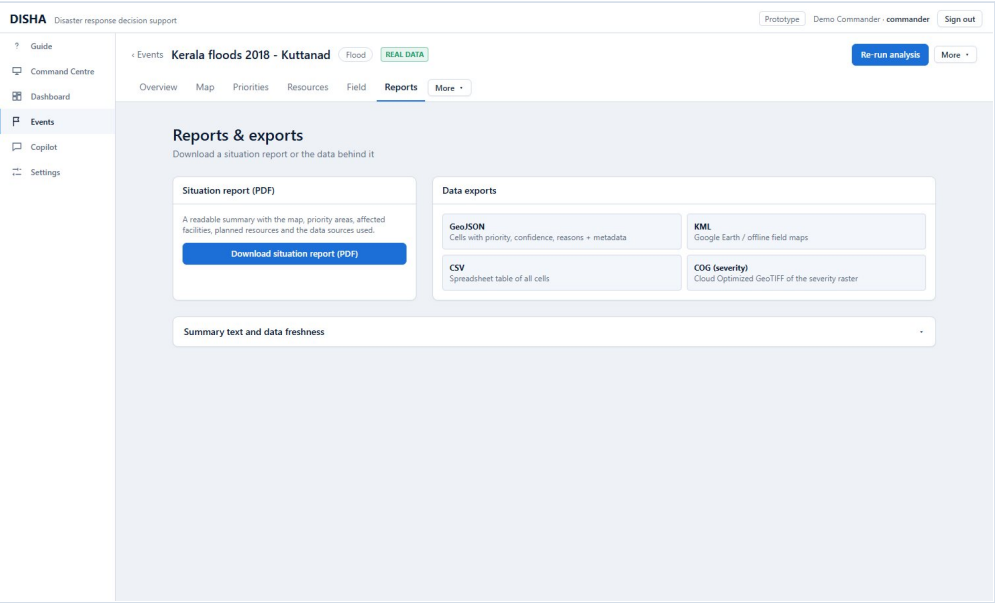


- Runs on the **simulated** Kerala scenario so it works without internet.
- Every step is narrated with the numbers it produced.
- Ends with a verified field report that changes a priority: the closed loop.
- Takes about 45 seconds; the browser test checks that it completes.

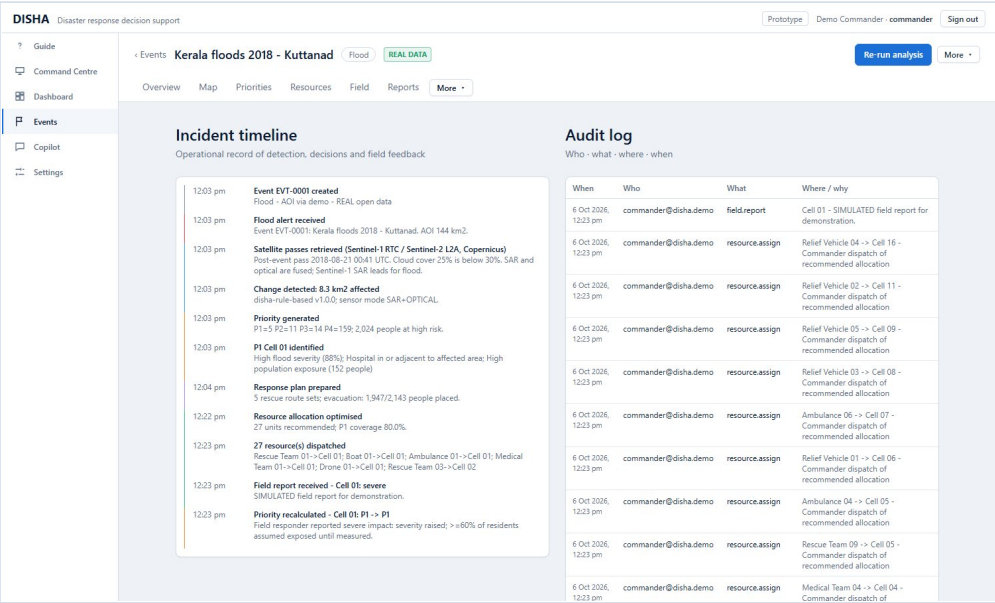
Alert, acquisition, detection, impact, priority, prediction, route, resources, field report and recalculated priority, narrated step by step.

Rest of the product

Reports, timeline and audit



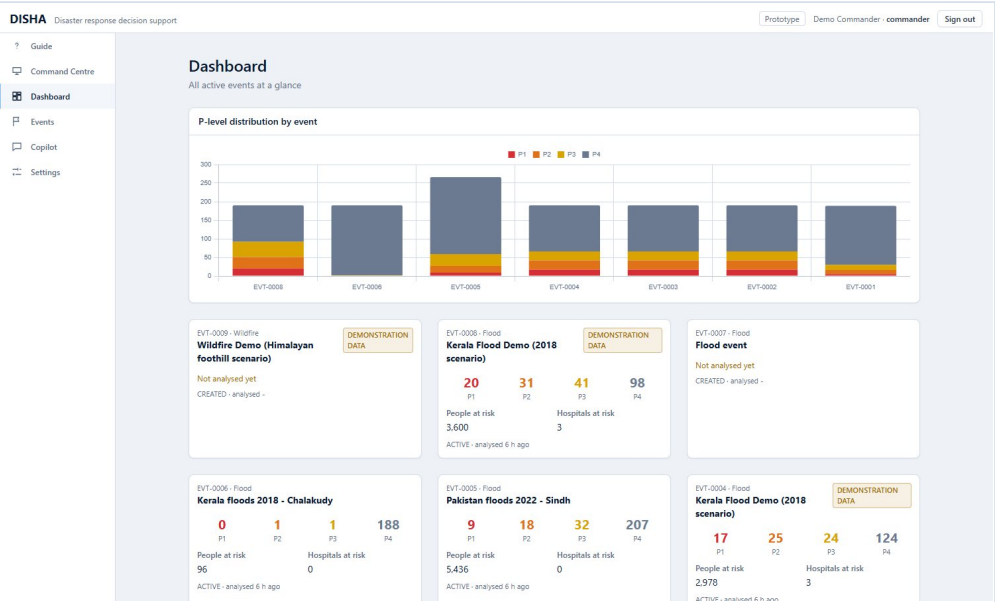
Reports: PDF situation report plus GeoJSON, KML, CSV and COG exports, each with event, time, sources, model and confidence metadata.



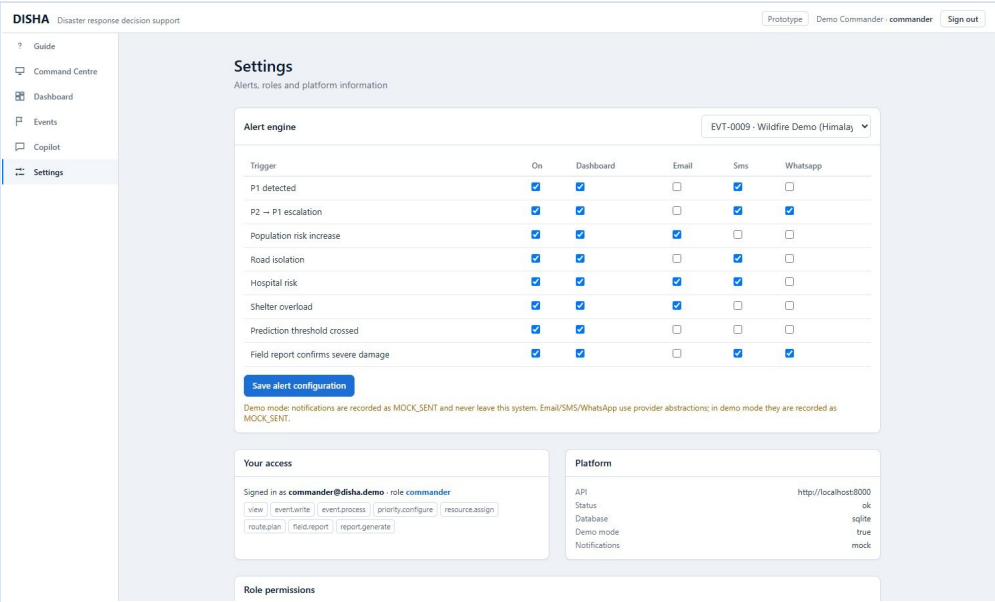
Incident timeline and audit log: who did what, where and when.

Rest of the product

Dashboard and settings



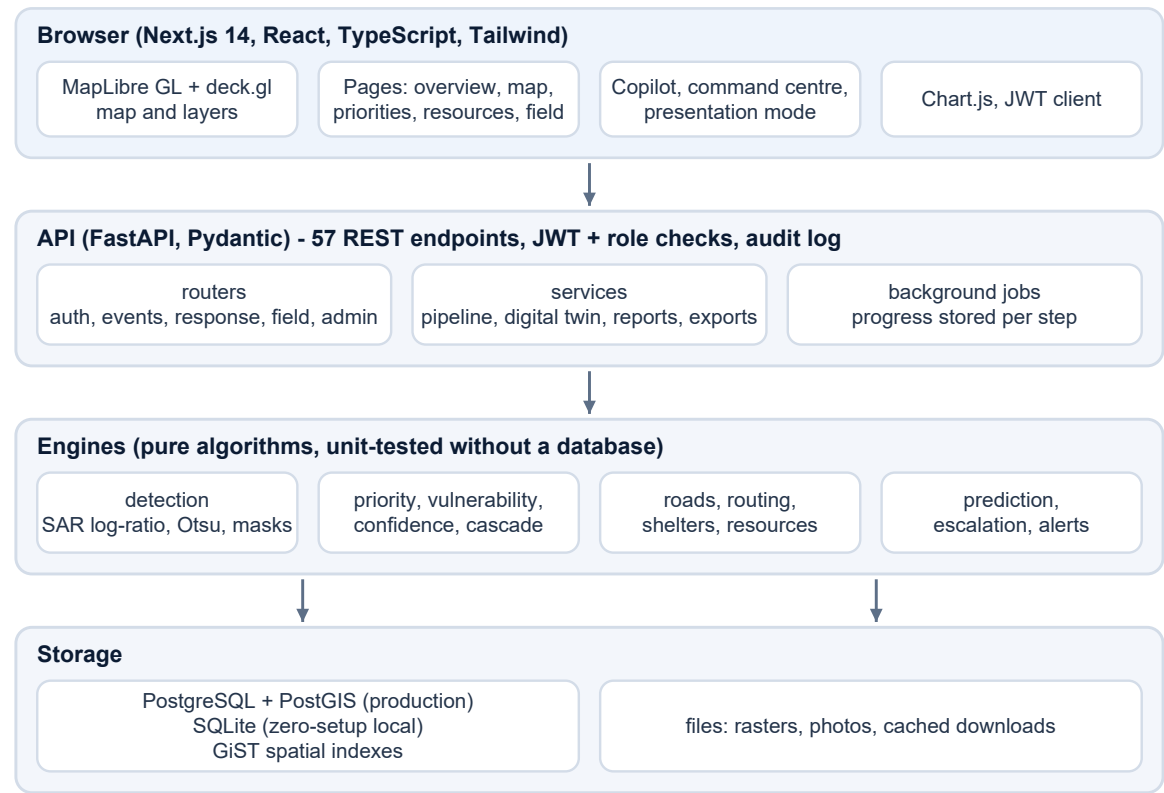
Dashboard: all events at a glance.



Settings: alert triggers and channels (email, SMS, WhatsApp are mock providers in this build), plus the role permission table.

Under the hood

Architecture



Technology

Layer	What is used
Frontend	Next.js 14, React, TypeScript, Tailwind CSS, MapLibre GL, deck.gl, Chart.js
Backend	Python, FastAPI, Pydantic, SQLAlchemy
Geospatial	H3, Shapely, GeoPandas, Rasterio, NetworkX, NumPy/SciPy
Database	PostgreSQL + PostGIS (GiST indexes), SQLite fallback
Reports	ReportLab (PDF), GeoJSON, KML, CSV, Cloud Optimised GeoTIFF
Quality	pytest, Playwright browser tests, GitHub Actions
Packaging	Docker, docker-compose (PostGIS, Redis, backend, frontend)

Design rule: the algorithms are plain functions that do not touch the database, so each one can be tested on its own. The same algorithms process real and simulated data.

Engineering

Tested and built in the open

35+

commits, built in logical steps

~7,000

lines of Python

~3,100

lines of TypeScript

53

automated tests (52 run in CI)

5

CI jobs, all passing

The screenshot displays the GitHub Actions interface for the repository `srushti117/Disha`. The left sidebar shows the 'Actions' tab with a list of workflow runs. The main area shows the 'All workflows' section with a search bar and a table of 8 workflow run results. The runs are as follows:

Workflow	Event	Status	Branch	Actor	Time
test(e2e): follow the renamed labels in the simplified UI	CI #17: Commit 4cbe3bc pushed by srushti117	Success	main	srushti117	22 minutes ago
fix(frontend): make the command-centre guide banner English	CI #16: Commit 6382ed6 pushed by srushti117	Success	main	srushti117	43 minutes ago
ci: make browser e2e and Docker image builds required; upd...	CI #15: Commit 403b0bb pushed by srushti117	Success	main	srushti117	Today at 4:10 PM
ci: remove Dependabot config	CI #14: Commit fb241cc pushed by srushti117	Success	main	srushti117	Today at 4:05 PM
github_actions in ./ - Update #1612753779	Dependabot Updates #3: by dependabot Bot	Success	main	dependabot Bot	Today at 3:59 PM
npm_and_yarn in /frontend - Update #1612753771	Dependabot Updates #2: by dependabot Bot	Success	main	dependabot Bot	Today at 3:59 PM
pip in /backend - Update #1612753767	Dependabot Updates #1: by dependabot Bot	Success	main	dependabot Bot	Today at 3:59 PM

GitHub Actions: backend tests (SQLite and PostGIS), frontend build, browser test, Docker image builds.

What is checked on every push

- Backend tests on SQLite **and** on a real PostGIS database.
- Frontend type-check and production build.
- A headless-Chrome run of the full workflow: login, guided demo, map, cell explanation, route, every page, copilot, role restrictions.
- Docker images build from a clean checkout.
- Tests for algorithms (priority formula, detection masks, routing around blocked roads, shelter capacity), the API, all scenarios, and the real-data code with the network mocked.

An opt-in test also runs the whole pipeline against the real data services.

Trust

Security, roles and accountability

Capability	Admin	Commander	Analyst	Responder	Observer
View everything	Yes	Yes	Yes	Yes	Yes
Create events, run analysis	Yes	Yes	Yes	-	-
Change priority weights	Yes	Yes	-	-	-
Plan and dispatch resources	Yes	Yes	-	-	-
Plan routes	Yes	Yes	-	Yes	-
Submit field reports	Yes	Yes	-	Yes	-
Generate reports	Yes	Yes	Yes	-	-
Manage users and data	Yes	-	Data only	-	-

Principles

- Sign-in with JWT tokens; passwords hashed with PBKDF2; every endpoint checks the role.
- Important actions (event changes, resource assignments, field reports, configuration, exports) are written to an audit log: who, what, where, why, when.
- Secrets live in environment variables; a template is provided and no secrets are in the repository.
- AI is decision support. Each important output shows its confidence, data sources, time and reasons; predictions are labelled estimates.
- Notifications (email, SMS, WhatsApp) use a provider interface; in this build they are mock and nothing is sent.
- The demo accounts and their password are public and are for demonstration only.

Honest assessment

What is not done, and what is assumed

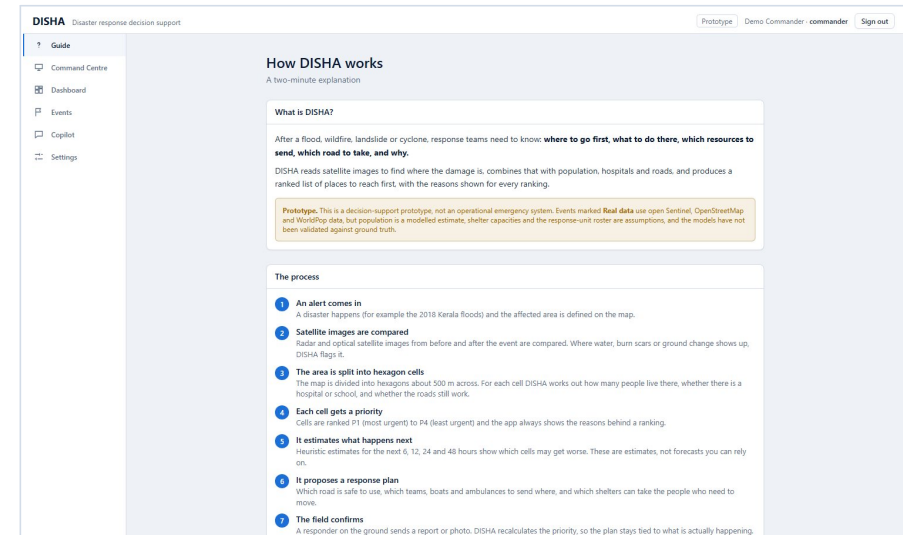
Area	Status
Detection accuracy on real events	Not validated against surveyed flood extents. No accuracy numbers are claimed; the app shows none unless real ground truth is registered.
Deep-learning models (Siamese U-Net, ChangeFormer)	Interfaces only; no trained weights are included. The rule-based detector is used and the fallback is reported.
Prediction model	Hand-set heuristic, uncalibrated. Always labelled as an estimate.
Population	Modelled estimate from WorldPop, not a census; children/elderly shares unavailable.
Shelters and rescue units	Capacities assumed by type; rescue roster hypothetical. Needs real registries before operational use.
Radar limits	Flooded built-up areas and flooded vegetation are under-detected; small landslides are not visible at 30 m.
Photo analysis	Simple colour/texture rules, not a trained model.
Notifications	Mock providers: nothing is actually sent.
Background jobs	Run inside the API process; Redis is provisioned but not yet used as a queue.
Deployment	Docker images build in CI; running the full docker-compose stack has not been exercised yet.
Live wildfire hotspots	Needs a free NASA FIRMS key; not exercised.

Next

Where this can go

Planned, not built. Items follow the original proposal and the gaps listed on the previous page.

- **Validate on real events** against surveyed flood extents; publish precision, recall and IoU.
- **Train deep-learning change models** on public datasets (xBD, Sen1Floods11, LEVIR-CD, Landslide4Sense, CaBuAr) and plug them into the existing detector interface.
- **More satellites:** ISRO EOS-04 and NISAR L-band radar, higher-resolution imagery for landslides.
- **Real registries** for shelters, rescue units and hospital capacity from state disaster authorities.
- **Real alerts:** SMS and WhatsApp through an approved provider, with delivery receipts.
- **Learning from the field:** use verified reports to retrain models (active learning).
- **Scale:** move jobs to a queue, serve imagery as map tiles, support many districts at once.
- **Pre-positioning:** use rainfall and cyclone-track forecasts to estimate impact before landfall.



The in-app guide explains the project in plain English.

For the judges

Questions you are likely to be asked

Question	Straight answer
Is this real satellite data?	Yes for events marked Real data: Sentinel-1/2, Copernicus DEM, WorldCover, WorldPop, OpenStreetMap and Open-Meteo are downloaded at run time. Simulated events are generated and labelled.
How accurate is it?	Unknown. The method is standard, but we have not checked it against surveyed flood maps, so we make no accuracy claim.
Why radar first?	Radar sees through cloud and at night, which matters most during monsoons and cyclones.
Why do the numbers differ from the slides?	Counts are computed from the data, not scripted. We set priority thresholds after looking at the score distribution; they are adjustable.
Is the AI making decisions?	No. It ranks and explains; commanders decide. Every output shows reasons, confidence and sources, and predictions are labelled estimates.
What is original here?	The step from change map to decision: priorities, explanations, routes, shelters, resources and a field-verification loop in one system.
What would it take to use this operationally?	Validation on real events, real resource and shelter registries, trained models, hardened security and a queue, as listed under limitations.
Can we see it run?	Yes: guided demo (about a minute, works offline) or a real-data event (first run downloads for 1 to 3 minutes).

Show it

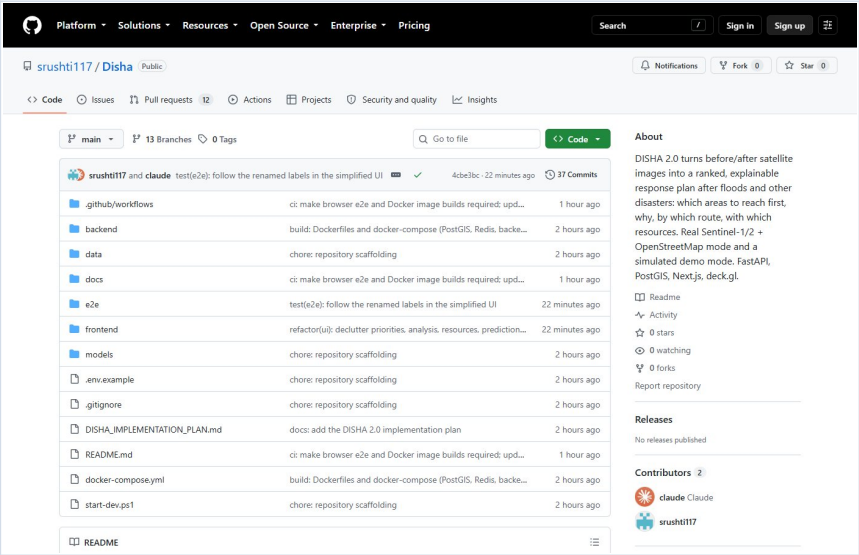
A three-minute demo script

- 0:00** Command Centre. Say: after a disaster, teams need to know where to go first and why.
- 0:20** Open *Events* and start **Kerala floods 2018 - Kuttanad (real data)**, then **Run analysis**. Mention it downloads real Sentinel images and OpenStreetMap roads (use a pre-run event if time is short).
- 0:50 Map**: switch on *Satellite image*, then show detected flood and priority hexagons. Click a hexagon: **why is it P1**.
- 1:30 Find safest route**, then **+6 h** on the time control (an estimate).
- 2:00 Resources**: Plan resources and Dispatch. Show shelters.
- 2:20 Field**: simulate a severe report. Show the priority change and *What changed*.
- 2:45 Reports**: download the situation report. Close with limits: radar blind spots, modelled population, unvalidated accuracy.

Run it yourself

Step	Command
Windows, no Docker	./start-dev.ps1 (API on :8000, UI on :3000)
Docker	cp .env.example .env, set secrets, then docker compose up --build
Open	http://localhost:3000
Demo login	commander@disha.demo / Disha@2026 (demo only)
Tests	cd backend; python -m pytest tests -q
Browser test	cd e2e; node run.mjs (stack running)

Roles to try: commander (everything), analyst (data, no dispatch), responder (missions, routes, field reports), observer (read-only), admin.



Public repository: github.com/srushti117/Disha

Reference

Glossary

Term	Meaning
Cell / hexagon	One H3 hexagon (about 500 m). The smallest unit DISHA scores. 'Cell 07' is hexagon number 7.
P1 to P4	Priority levels: Critical, High, Moderate, Low.
Priority score	0 to 1: 35% severity, 30% people, 20% facilities, 15% road access lost.
SAR / radar	Satellite radar (Sentinel-1). Sees through cloud and at night. Water looks dark.
Log-ratio	After minus before brightness in dB; a large drop indicates new water.
Otsu threshold	A standard way to pick a cut-off between two groups of pixel values automatically.
Sentinel-2	Optical (colour) satellite images, blocked by cloud.
DEM / slope	Elevation model; slope removes places water cannot pool.
WorldPop / WorldCover	Open population and land-cover datasets used to estimate people per area.
OpenStreetMap	Open map of roads, hospitals, schools and place names.
Digital twin	The stored live state of each hexagon, recalculated when new information arrives.
Confidence	How much DISHA trusts a result: high, medium, low or insufficient data.
Estimate	A prediction, always shown with a probability and never as a certainty.
Field verified	A responder confirmed the situation; this can change a priority.
Real data / Simulated	Whether an event used actual open data or generated demonstration data.